

ORGANIC CHEMISTRY – CRACKING & ALKENES

KEY VOCABULARY

Cracking: breaking long hydrocarbons into shorter, useful ones.

Thermal decomposition: breaking down using heat.

Alkene: an unsaturated hydrocarbon with a C=C bond.

Catalytic cracking: uses a hot catalyst to crack vapour.

COMMON MISCONCEPTIONS

~~Cracking makes only one product.~~

✓ It makes a shorter alkane AND an alkene.

~~Cracking creates atoms.~~

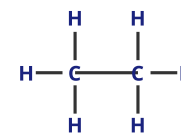
✓ It just rearranges the same atoms into smaller molecules.

~~Alkenes are saturated.~~

✓ Alkenes are unsaturated (have a C=C).

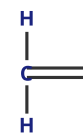
ALKANE VS ALKENE

ETHANE (alkane)



C₂H₆ · saturated

ETHENE (alkene)



C₂H₄ · unsaturated (C=C)

1 DEFINE CRACKING

State what is meant by **cracking**. [1 mark]

2 WHY CRACK

Explain why long-chain hydrocarbons are cracked. [2 marks]

3 TEST FOR ALKENE

Describe the **bromine water test** and its result for an alkene. [2 marks]

4 BALANCE CRACKING

Complete: C₁₀H₂₂ → C₈H₁₈ + ____ . [1 mark]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Cracking produces a shorter ____ and an ____ that is unsaturated.

alkene

alkane

saturated

unsaturated

6 USES OF ALKENES

Give one use of alkenes. [1 mark]